Inguinal Hernia and Hydrocele

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Definitions

- ☑ Inguinal hernia is protrusion of intra-abdominal structures, such as omentum, bowel, or ovary through the open processus vaginalis into the inguinal canal or scrotum.
- ☑ Most inguinal hernias in children are indirect; direct hernias and femoral hernias are very rare.
- ☑ Hydrocele is collection of fluid inside the tunica vaginalis in the scrotum or along the spermatic cord. They can be communicating (most common) or encysted or rarely abdominoscrotal hydroceles.

Etiology/Epidemiology

- ☑ Patency (nonobliteration) of patent processus vaginalis (PPV) is the cause of hernia and hydrocele; a wide patency results in inguinal hernia, while a narrow PPV may cause hydrocele.
- ☑ Secondary hydroceles are less common, and causes include trauma, infection, or tumors.
- ☑ Hydrocele is seen in about 6% of infants; spontaneous resolution occurs in most cases by 18 months. Inguinal hernia occurs in 1–5% of children, more common in preterm babies (up to 30%). Hernia is more common in boys than girls.

Clinical Presentation, Diagnosis, and Investigations

- ☑ Inguinal hernia typically presents as a painless, intermittent swelling in the groin, appearing with straining/crying. Pain is a feature of irreducible or obstructed hernia.
- ☑ Hydrocele also presents as a painless swelling in the scrotum with a gradual waxing/waning of size but no sudden increase/decrease of size.
- ☑ Diagnosis of inguinal hernia and hydrocele is clinical, based on history and clinical examination. If the hernia cannot be seen by the doctor in the clinic (as happens commonly), the parents can be asked to take a picture of the swelling on their mobile phone, which can be diagnostic. The diagnosis of hydrocele is more straightforward. Testis is palpable within the hydrocele (except in tense hydroceles); testis is palpable separately below the swelling in encysted hydroceles.
- ☑ Ultrasound is only required if a tense hydrocele needs to be differentiated from an incarcerated hernia, a suspected abdominoscrotal hydrocele or to look for ovary inside a female hernia.
- ☑ Routine laboratories examination may only be required to assess fitness for anesthesia and surgery.
- ☑ Differential diagnosis of inguinal hernia may be hydrocele, inguinal lymphadenitis, inguinoscrotal cellulitis or abscesses, varicocele, testicular tumor, and undescended testis.
- ☑ Mucocele/pneumatocele in early neonatal life may be an important differential diagnosis in patients with history or clinical findings pointing toward necrotizing enterocolitis (NEC) or perforation peritonitis.

Management

- ☑ Inguinal hernia in children always needs surgery (herniotomy) which involves the division and ligation of PPV at the internal inguinal ring. An early surgery is recommended once the diagnosis is confirmed. The procedure can be done by open (groin incision) and laparoscopic (intracorporeal or extracorporeal) techniques.
- ☑ Laparoscopy has the specific advantages of minimal handling of vas and vessels and the opportunity to observe the presence of contralateral PPV. If present, the contralateral PPV can be closed in the same sitting.
- ☑ Up to one-third of boys and over half of girls with unilateral inguinal hernia reveal a contralateral PPV on laparoscopy. In girls with hernia, laparoscopy also offers the chance to visualize the internal genitalia and diagnoses the uncommon but important condition of complete androgen insensitivity syndrome (CAIS).
- ☑ In children under 18 months of age, hydroceles can be simply observed, especially if they are small and lax. Large and tense hydroceles may require surgery, especially in an older child. Surgery for hydrocele also involves the division and closure of PPV, along with removal of fluid from the distal sac.

Management

Inguinal Hernia—Special Situations

- ☑ Children presenting as emergency with irreducible/incarcerated hernia should have attempted reduction of hernia under sedation. Once reduced, elective herniotomy may be performed typically in 24–48 hours, by which time the swelling would have reduced. If the hernia cannot be reduced even under sedation, urgent exploration is required. Complications (recurrence and testicular atrophy) of emergency herniotomy for irreducible hernia are higher than elective herniotomy.
- ☑ Preterm babies with hernia should be preferably operated under caudal block and sedation without intubation, to reduce the risks of general anesthesia with endotracheal intubation.
- ☑ Children with inguinal hernia and associated undescended testis should be assessed intraoperatively after herniotomy. If good cord length is easily achievable, then orchidopexy can be performed at the same setting. However, if cord length is tight, a staged procedure with later orchidopexy may ensure better success.
- ☑ As mentioned earlier, inguinal hernia in girls may be the presentation of CAIS. On clinical examination, testis may be felt as content of the inguinal hernia. However, the diagnosis is most often made during the surgery. Herniotomy may be then performed after counseling the family (a gonadal biopsy may be required). Karyotyping confirms the diagnosis. The gonads may be removed after natural puberty is attained.
- ☑ During open herniotomy for unilateral hernia, contralateral herniotomy is not recommended if contralateral side is clinically normal.
- ☑ In children with prune belly syndrome or bladder exstrophy, large defect at hernia site may need hernioplasty in addition to ligation of sac.

Surgical Outcome and Complications

- ☑ Children with unilateral hernia have about 10% risk of later development of inguinal hernia on the contralateral side. This risk may be underestimated due to inadequate follow-up. This risk may be higher, especially in preterm and very low birth weight (VLBW) children.
- ☑ Recurrence of inguinal hernia after surgery occurs in 1% (0.7–3.8%) of children. Recurrence is more common in premature babies, postincarceration, and with associated conditions such as ventriculoperitoneal (VP) shunt, exstrophy, or connective tissue disorders. Laparoscopy is the treatment of choice for recurrent inguinal hernias in children.
- ☑ Other complications include distal hydrocele, testicular ascent (acquired undescended testis), and testicular atrophy (especially with irreducible hernias).

Further Reading

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